

Senior Data Scientist leading analytics at a **Harvard Innovation Labs** startup, building ML, simulation models and the production backends that serve them, including climate and infrastructure modelling for cities. Research on clean-tech cost curves at Oxford & UCL. Published work on cities & innovation economies. Economist by training & ex-Baringa consultant.

TECHNICAL SKILLS

Languages & ML | Python, R, SQL, JavaScript (scikit-learn, XGBoost, PyTorch, PyTorch Geometric, NetworkX, SciPy)
Geospatial | GeoPandas, Shapely, Rasterio, xarray, PySAL, OSMnx, momepy, H3, PostGIS, QGIS, EarthEngine, CARTO
Data Engineering & Cloud | AWS, GCP, Supabase (PostgreSQL), Snowflake, REST APIs, GitHub, D3.js, Claude Code
Methods | Statistical & Econometric Modelling, Machine Learning, Forecasting, Monte Carlo & Agent-Based Simulation, Network Science, Spatial Statistics, Remote Sensing, Imputation & Data Validation

PROFESSIONAL EXPERIENCE

Aretian Urban Analytics | Senior Data Scientist & Software Developer | London, UK **Nov 2025 – Present**

Leading data science. Advising governments, utilities, investors & developers on how cities grow. Manage 2 junior data scientists own projects end to end, from framing the client's question to production models & presenting results to leadership.

Applied science for cities & infrastructure: statistical, ML and simulation models on climate, energy, housing and growth for utilities, governments, investors and developers across Europe, the US, Africa and the Middle East

- Modelled heat and flood risk across 216,000 neighbourhood-scale areas of Barcelona, validated against 15 years of mortality data; costed a €100M adaptation programme and built a Monte Carlo appraisal of lives saved per euro
- Forecast energy demand across Spain to show where new grid capacity and infrastructure investment will be needed
- Benchmarked 12 city economies, including Saudi Arabia's NEOM region, as industry networks built from 500K+ business records per city, to identify growth sectors and inform master planning and investment priorities
- Clustered 2,920 high-risk areas into six neighbourhood types (k-means) and built a cooling model to match each type with the most cost-effective intervention, finding trees ~25% cheaper per °C of cooling than coatings

Data platforms & engineering: built much of the backend behind Aretian's products, including cdt.cat, a live digital twin of Catalonia that municipalities, investors and real estate firms use for planning and investment

- Engineered the cdt.cat backend on AWS and Supabase, joining 2.9M property records with housing, economic and infrastructure data into one queryable dataset, plus a simulation of where 641,000 new homes could be built
- Unified messy data from many sources into clean pipelines: rate-limited API harvesters (170,000+ publications), business registries in nine formats across 12 cities, and LLM-assisted extraction from unstructured reports
- Designed regression-based imputation and validation checks that raised data coverage from under 45% to 97%+ and caught a pipeline error overstating population (13M vs 7.8M); set team standards for reproducible models

University of Oxford | Inst. for New Economic Thinking | Volunteer Researcher | Oxford, UK **Apr 2026 – Sep 2026**

Forecasting Technological Change Programme, estimating how fast clean-tech costs fall w/ François Lafond & Amir Akhter

- Estimated Wright's law experience curves for air conditioning: costs fall ~30–35% per doubling of production when measured per kW of cooling, against ~4% on raw prices; tested robustness across five efficiency scenarios
- Compiled a 1952–2026 dataset of 26 price points from primary sources, every point's source & transformation logged
- Showed official US producer price statistics miss most of the real cost decline (–6% vs –57% per kW, 1989–2019)

Norman Foster Institute | Data Scientist | Madrid, Spain

Jan 2025 – Dec 2025

- Given a €52,000 research grant to apply spatial data science to sustainable development in Cape Town, South Africa
- Built a remote-sensing densification model in Google Earth Engine combining LIDAR building heights, zoning,

housing capacity and population density to find where Cape Town has room for new housing and economic growth

- Designed a data-driven tool scoring the quality of Cape Town's public spaces against the city's development plan

Space Syntax Ltd. | Data Science Researcher | London, UK

Apr 2024 – Sep 2024

- Built a Python pipeline analysing the UK's entire road network (~4m road segments) alongside Ordnance Survey business location data, using high-performance graph software to make a national-scale analysis feasible
- Used ML (clustering) to group 534 towns and cities into 11 types based on the shape and layout of their street networks
- Tested a common urban planning assumption that shops concentrate on the best-connected streets; found street layout explained only ~11% of where shops are located, while shops clustered strongly together, pointing to other factors
- Offered a full-time Spatial Data Scientist role on completion

Baringa Partners | Management Consultant (Data & Analytics) | London, UK

Mar 2021 – Aug 2023

Advised senior leadership on cost, technology and AI decisions for clients including Vodafone, Freshfields, Thomson Reuters and Penguin Random House, building the models and data pipelines behind their decisions as project manager

- Built two predictive data pipelines for Freshfields (Magic Circle law firm) estimating the probability of winning a dispute, expected legal costs and likely payout, as part of the firm's AI strategy
- Delivered a ~£10m opex reduction by redesigning Vodafone UK's consumer division ahead of a merger, and designed its Data & Analytics team with senior leadership
- Led a data pipeline build for the UK's largest music licensing company, increasing ingestion speed by 500%

Q5 Partners | Management Consultant (Intern) | London, UK

Sep 2020 – Mar 2021

- Supported a restructuring and organisational design project for National Grid (West of England)

Boston Consulting Group | Mobility Analyst | London, UK

Sep 2018 – Oct 2019

- Researched autonomous vehicles, smart cities and infrastructure as a student researcher

EDUCATION

M.Sc Spatial Data Science | University College London | Distinction

2023 – 2024

- Specialised in applying complexity science and ML to geospatial problems at *The Bartlett School of Built Environment*
- Awarded funding from *Space Syntax Ltd.* and the *Consumer Data Research Centre* to write MSc dissertation

M.A (Hons) Economics | University of Edinburgh | First Class

2016 – 2020

ACHIEVEMENTS

- 3rd Place | Society for Location Analysis *UK Masters Dissertation Awards* | 11/2024
- 1st Place | Centre for Advanced Spatial Analysis Hackathon w/ West Midlands Fire Service | £2000 Cash Prize | 06/2024
- 1st Place | Boston Consulting Group *London University Case Study Competition* | £500 Cash Prize | 09/2019

PUBLICATIONS & PROJECTS

- Norman Foster Institute (2026), *Research on Cities: Volume II*. Ivory Press & Norman Foster Foundation. Available here: <https://ivorypress.com/en/product/research-on-cities-volume-ii/>.
- Gras, R., Desai, N., Magini, F., de La Ville-Baugé, G., Munn, E., Agrawal, R. and Ruess, D. (2026). *A Vision for Boston's Innovation Economy: City Science Illuminating Urban Design, Economic, and Innovation Strategies*. Executive Summary Report. Cambridge, MA: Aretian Urban Analytics and Design.
- **Solar Valuation** | App valuing a solar project by simulating 10,000 scenarios and their probabilities
- **La Rambla Climate ABM** | Agent-based simulation of pedestrian heat exposure before and after a shade intervention
- **Harvard & MIT Research Networks** | Interactive tool: Co-authorship networks across Harvard & MIT, showing how collaboration drives research output